Lean decide certificate for the Nagao 2015/984 Lemma 4 counterexample

2026-10-29. Source finding KN-FIND-936151 (write-once, not edited). Full text: report.md.

Scope. A kernel-checked finite computation over F_2, $N \le 3$. It asserts nothing about Nagao's Proposition 5, Semaev's Assumption 1, $p \ge 3$, $N > 3$, the literal >= reading of condition (1), or any elliptic curve. It is no stronger than the Python certificates it mirrors (tools/lemma4_inside_form.py, recheck.py).

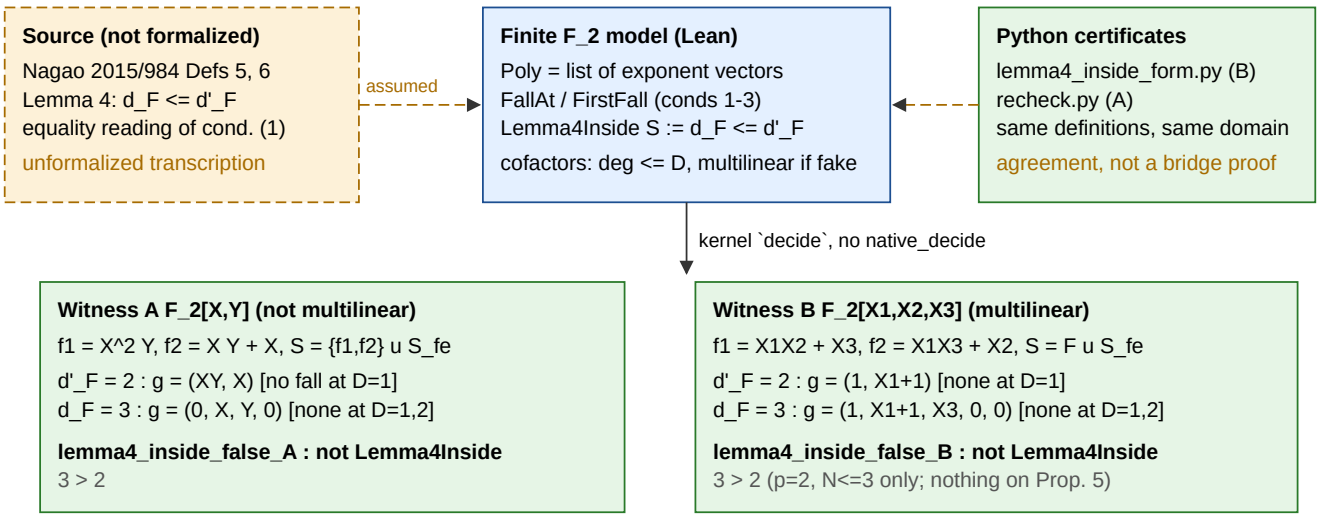


Figure 1: What is proved (green), assumed (dashed), and mirrored.

Theorem. Equality reading of condition (1), field equations $X_i^2 + X_i$ as generators of the true system (inside form), $d_F(S) \le d'_F(S)$ for $S = \{f_1, f_2\} \cup S_fe$:

witness	ring	f_1, f_2	d'_F / d_F	Lean theorem
A	$F_2[X,Y]$	X^2Y, $XY + X$	2 / 3	lemma4_inside_false_A
B	$F_2[X_1,X_2,X_3]$	$X_1X_2 + X_3$, $X_1X_3 + X_2$	2 / 3	lemma4_inside_false_B

Corollary lemma4_inside_false. All goals by decide +kernel; axioms: propext, Classical.choice, Quot.sound. Receipt verification.json: machine_verified, semantic review pending.

Hypothesis-delta table (declaration lemma4_inside_false):

binder / side condition	claim states it?	status
Poly = exponent-vector list, odd multiplicities	over F_2	implied (no bridge to MvPolynomial)
equality reading of cond. (1)	finding §3	implied; >= reading out of scope
cofactors: degree ≤ D, multilinear if fake	no	**scoping**: leading-form argument not in Lean
fake side drops generators reducing to 0	finding §2	**scoping**: only $X_i^2+X_i \to 0$ checked
FirstFall over D ≥ 1, sum ≠ 0	Defs 5/6	implied

Two scoping rows cap fidelity at *partial*. Unformalized step (proof_map.prerequisites): the transcription of Nagao's Definitions 5/6 into the finite model. Not added: independence from the chosen definitions, the outside form, any consumer's instances.